

Lecture 17

Metropolis-Hastings sampling

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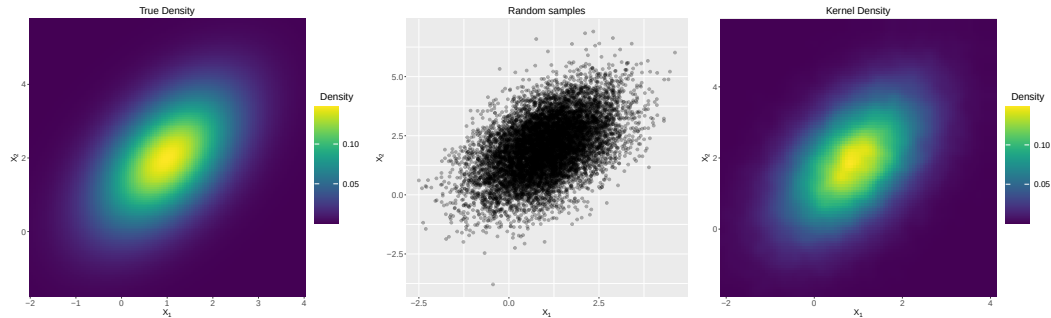
(Edited from the lecture notes by Brian J. Reich, NC State)



Recap: Monte Carlo sampling

- ▶ Monte Carlo (MC) sampling is the predominant method of Bayesian inference.
- ▶ The main reason is that it generally performs better than other alternatives for high-dimensional models (i.e., with many parameters).
- ▶ The main idea is to approximate posterior summaries in terms of some randomly drawn samples from the posterior distribution.
- ▶ This often requires drawing samples from non-standard distributions.
- ▶ It also requires careful analysis to be sure the approximation is sufficiently accurate.

Example



Recap: Gibbs sampling

A Set initial value $\boldsymbol{\theta}^{(0)} = (\theta_1^{(0)}, \dots, \theta_p^{(0)})$

B For iteration t ,

FC1 Draw $\theta_1^{(t)} | \theta_2^{(t-1)}, \dots, \theta_p^{(t-1)}, \mathbf{Y}$

FC2 Draw $\theta_2^{(t)} | \theta_1^{(t)}, \theta_3^{(t-1)}, \dots, \theta_p^{(t-1)}, \mathbf{Y}$

...

FCp Draw $\theta_p^{(t)} | \theta_1^{(t)}, \dots, \theta_{p-1}^{(t)}, \mathbf{Y}$

We repeat step B S times and get the posterior draws

$$\boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(S)}.$$

Recap: Gibbs sampling for the Gaussian model

- ▶ Say $Y_i \stackrel{iid}{\sim} \text{Normal}(\mu, \sigma^2)$ with $\mu \sim \text{Normal}(0, \sigma_0^2)$, and $\sigma^2 \sim \text{InvGamma}(a, b)$.
- ▶ In Chapter 2 (while discussing conjugate priors), we saw that if we knew either μ or σ^2 , we can sample from the other parameter.
- ▶ $\mu | \sigma^2, \mathbf{Y} \sim \text{Normal} \left(\frac{n\bar{Y}\sigma^{-2} + \mu_0\sigma_0^{-2}}{n\sigma^{-2} + \sigma_0^{-2}}, \frac{1}{n\sigma^{-2} + \sigma_0^{-2}} \right)$.
- ▶ $\sigma^2 | \mu, \mathbf{Y} \sim \text{Inverse-Gamma} \left(\frac{n}{2} + a, \frac{1}{2} \sum_{i=1}^n (Y_i - \mu)^2 + b \right)$.
- ▶ But how to draw from the joint distribution $\pi(\mu, \sigma^2 | \mathbf{Y})$? Using Gibbs sampling.

Metropolis sampling

- ▶ In Gibbs sampling, each parameter is updated by sampling from its full conditional distribution.
- ▶ This is only possible with conjugate priors.
- ▶ However, if the prior is not conjugate, it is not obvious how to make a draw from the full conditional.
- ▶ For example, if $Y \sim \text{Normal}(\mu, 1)$ and $\mu \sim \text{Beta}(a, b)$ then

$$p(\mu|Y) \propto \exp\left[-\frac{1}{2}(Y - \mu)^2\right] \mu^{(a-1)}(1 - \mu)^{b-1}.$$

- ▶ For some likelihoods, there is no known conjugate prior, e.g., logistic regression.
- ▶ In these cases, we use Metropolis sampling.

Metropolis sampling

- ▶ Metropolis sampling is a version of rejection sampling.
- ▶ Let θ_j^* be the current value of the parameter being updated and $\theta_{(j)}$ be the current value of all other parameters.

- ▶ You propose a random candidate based on the current value, e.g.,

$$\theta_j^c \sim \text{Normal}(\theta_j^*, s_j^2).$$

- ▶ The candidate is accepted with probability

$$R = \min \left\{ 1, \frac{p(\theta_j^c | \theta_{(j)}, \mathbf{Y})}{p(\theta_j^* | \theta_{(j)}, \mathbf{Y})} \right\}.$$

- ▶ If the candidate is not accepted then you simply retain the previous value and move to the next step.

Metropolis sampling

- ▶ The candidate standard deviation s_j is a tuning parameter.
- ▶ Ideally s_j is tuned to give acceptance probability around 0.3–0.4.
- ▶ If s_j is too small: Acceptance rate is large but the algorithm fails to explore the posterior well.
- ▶ If s_j is too large: Acceptance rate is poor while the algorithm again fails to explore the main important region of the posterior well.
- ▶ Off-the-shelf programs have default values, and many allow you to change the value if the results are unsatisfactory.

Metropolis-Hastings sampling

- ▶ Denote $\theta_j^c \sim q(\theta|\theta^*)$ as the candidate distribution.

- ▶ The candidate distribution is symmetric if

$$q(\theta^*|\theta_j^c) = q(\theta_j^c|\theta^*).$$

- ▶ For example, if $\theta_j^c \sim \text{Normal}(\theta_j^*, s_j^2)$ then

$$q(\theta_j^c|\theta^*) = \frac{1}{\sqrt{2\pi}s_j} \exp \left[-\frac{(\theta_j^c - \theta_j^*)^2}{2s_j^2} \right] = q(\theta^*|\theta_j^c).$$

Metropolis-Hastings sampling

- ▶ Metropolis-Hastings (MH) sampling generalizes Metropolis sampling to allow for asymmetric candidate distributions.
- ▶ For example, if $\theta_j \in [0, 1]$ then a reasonable candidate is

$$\theta_j^c | \theta_j^* \sim \text{Beta}[10\theta_j^*, 10(1 - \theta_j^*)].$$

- ▶ Then $q(\theta_j^* | \theta_j^c)$ and $q(\theta_j^c | \theta_j^*)$ are both beta PDFs.
- ▶ MH proceeds exactly like Metropolis except the acceptance probability is

$$R = \min \left\{ 1, \frac{p(\theta_j^c | \theta_{(j)}, \mathbf{Y}) q(\theta_j^* | \theta_j^c)}{p(\theta_j^* | \theta_{(j)}, \mathbf{Y}) q(\theta_j^c | \theta_j^*)} \right\}.$$

Metropolis-Hastings sampling

- ▶ What if we take the candidate distribution to be the full conditional distribution

$$\theta_j^c \sim p(\theta_j^c | \theta_{(j)}, \mathbf{Y})?$$

- ▶ What is the acceptance ratio?

$$\frac{p(\theta_j^c | \theta_{(j)}, \mathbf{Y}) q(\theta_j^* | \theta_j^c)}{p(\theta_j^* | \theta_{(j)}, \mathbf{Y}) q(\theta_j^c | \theta_j^*)} = \frac{p(\theta_j^c | \theta_{(j)}, \mathbf{Y}) p(\theta_j^* | \theta_{(j)}, \mathbf{Y})}{p(\theta_j^* | \theta_{(j)}, \mathbf{Y}) p(\theta_j^c | \theta_{(j)}, \mathbf{Y})} = 1.$$

- ▶ What does this say about the relationship between Gibbs and Metropolis Hastings sampling?
- ▶ Gibbs sampling is a special case of the Metropolis-Hastings sampling where the candidate distribution is same as the full conditional distribution.

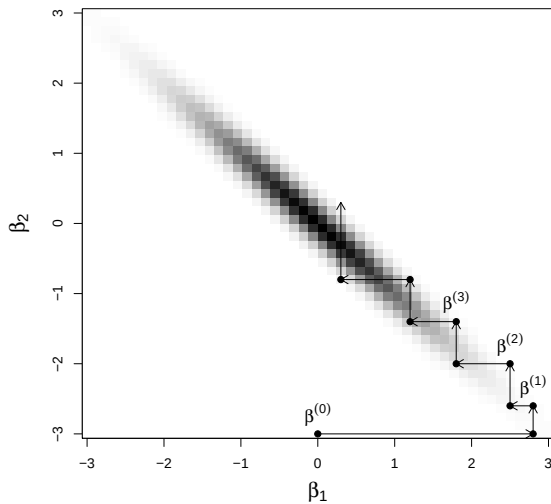
Variants

- ▶ You can combine Gibbs and Metropolis in the obvious way, sampling directly from full conditional when possible and Metropolis otherwise.
- ▶ Adaptive MCMC varies the candidate distribution throughout the chain.
- ▶ Hamiltonian MCMC uses the gradient of the posterior in the candidate distribution and is used in STAN.

Blocked Gibbs/Metropolis

- ▶ If a group of parameters are highly correlated, convergence can be slow.
- ▶ One way to improve Gibbs sampling is a block update.
- ▶ For example, in linear regression might iterate between sampling the block $(\beta_1, \dots, \beta_p)$ and σ^2 .
- ▶ Blocked Metropolis is possible too.
- ▶ For example, the candidate for $(\beta_1, \dots, \beta_p)$ could be a multivariate normal.

Posterior correlation leads to slow convergence



Summary

- ▶ With the combination of Gibbs and Metropolis-Hastings sampling, we can fit virtually any model.
- ▶ In some cases, Bayesian computing is actually preferable to maximum likelihood analysis.
- ▶ In most cases Bayesian computing is slower.
- ▶ However, in the opinion of many it is worth the wait for improved uncertainty quantification and interpretability.
- ▶ In all cases, it is important to carefully monitor convergence.

Thank you!